Complete at least 2 exercises.

Throughout, $\Bbbk$ is a field, and all modules are right modules unless a left action is specified.

Exercise 1.

(1) Let $A = \Bbbk A_n$ be the algebra of upper triangular $n \times n$ matrices with $n \geq 3$, graded by $\deg e_{ij} = j - i$. Describe the minimal graded projective resolution of each simple module L_i, $1 \leq i \leq n$, and show that it is linear. Deduce that A is Koszul.

(2) Let $S = \bigoplus_{i=1}^{r} \Bbbk e_i$, let V be a finite-dimensional S-bimodule in degree 1, and put

$$C_{ij} = \dim_{\Bbbk} e_i V e_j, \qquad A = T_S(V)/(V \otimes_S V) = S \oplus V.$$

Prove that

$$A^! = T_S(D(V)), \qquad h_A(t) = I + tC, \qquad h_{A^!}(t) = (I - tC^{\mathsf{T}})^{-1}.$$

(3) Let $S = \bigoplus_{i=1}^{4} \Bbbk e_i$ with coordinate idempotents e_i, and let $V = \Bbbk a \oplus \Bbbk b \oplus \Bbbk c \oplus \Bbbk d$ be the S-bimodule with

$$a \in e_1 V e_2, \qquad b \in e_2 V e_4, \qquad c \in e_1 V e_3, \qquad d \in e_3 V e_4,$$

and $e_i V e_j = 0$ for all other (i, j). For $\lambda \in \Bbbk$ put $A_\lambda = T_S(V)/(ab - \lambda cd)$. Find $R^\perp$ and a presentation of $A_\lambda^!$, for every $\lambda \in \Bbbk$.

(4) With A_λ as in (3), prove that $A_\lambda \cong A_\mu$ as graded algebras if and only if λ, μ are either both zero or both non-zero.

Hint. A graded isomorphism over S sends each of a, b, c, d to a scalar multiple of itself.

(5) Let $S = \bigoplus_{i=1}^{4} \Bbbk e_i$ and let $V = \Bbbk a \oplus \Bbbk b \oplus \Bbbk c$ be the S-bimodule with $a \in e_1 V e_2$, $b \in e_2 V e_3$, $c \in e_3 V e_4$ and $e_i V e_j = 0$ for all other (i, j). Let $R = \Bbbk\,(a \otimes b) \subset V \otimes_S V$ and $A = T_S(V)/(R)$. Find a basis of A and $h_A(t)$. Compute $R^\perp$, a presentation and a basis of $A^!$, and $h_{A^!}(t)$, and verify

$$h_A(t)\, h_{A^!}(-t)^{\mathsf{T}} = I_4.$$

Exercise 2.

(1) Let $B = T_{\Bbbk}(\Bbbk x \oplus \Bbbk y)/(x^2)$ with $\deg x = \deg y = 1$, and $a_d = \dim_{\Bbbk} B_d$. Prove that $a_d = a_{d-1} + a_{d-2}$ for $d \geq 2$ and $h_B(t) = \dfrac{1+t}{1-t-t^2}$. Then compute $B^!$ (with $X = x^*$, $Y = y^* \in D(V)$), a homogeneous basis of $B^!$ and $h_{B^!}(t)$, and verify $h_B(t)\, h_{B^!}(-t) = 1$.

(2) For $q \in \Bbbk$ let

$$C_q = \Bbbk\langle x, y\rangle/(x^2,\ y^2,\ xy + qyx), \qquad \deg x = \deg y = 1.$$

Find a homogeneous basis of C_q (justify linear independence) and determine, for each q, the set

$$\{(\alpha, \beta) \in \Bbbk^2 \mid u^2 = 0 \ \text{ for } u = \alpha x + \beta y\}.$$

(3) With C_q as in (2), prove that

$$C_q \cong C_{q'} \ \text{ as graded algebras} \quad \Longleftrightarrow \quad q = q' \ \text{ or } \ qq' = 1.$$

Deduce that $C_1 \ncong C_{-1}$ if $\operatorname{char} \Bbbk \neq 2$ (even ungraded), and say what changes if $\operatorname{char} \Bbbk = 2$.
Hint. A graded isomorphism is determined by a linear bijection of the degree-1 parts, and preserves $\{\Bbbk u \mid u^2 = 0\}$; you may use that $\{1, x, y, yx\}$ is a basis of C_q and $u^2 = \alpha\beta(1-q)\, yx$ for $u = \alpha x + \beta y$.

(4) For C_q as in (2), compute a quadratic presentation of $C_q^!$ (with $X = x^*$, $Y = y^* \in D(V)$) and $h_{C_q^!}(t)$. Compare $C_0^!$ with $C_1^!$.

Exercise 3.

(1) Let $A = \Bbbk[x]$ with $\deg x = 1$. Prove that $\operatorname{rad} A = 0$ whereas $A_+ = (x) \neq 0$.

(2) Let $A = \Bbbk[u, v]/(uv)$ with $\deg u = \deg v = 1$. Show that the minimal graded projective resolution of $\Bbbk = A/A_+$ has $P_0 = A$ and $P_h \cong A(-h)^{\oplus 2}$ for all $h \geq 1$.
Hint. Show that $ua + vb = 0$ in A if and only if $a \in vA$ and $b \in uA$.

(3) Let A be as in Lecture 5, let M be a bounded below locally finite-dimensional graded right A-module with minimal graded projective resolution $P_\bullet \to M$, and let $n \geq 0$. Prove that the following are equivalent:
   (a) $P_h = 0$ for all $h > n$;
   (b) $\operatorname{ext}_A^{n+1}(M, L(j)) = 0$ for every simple graded S-module L in degree 0 and every $j \in \mathbb{Z}$.
   Hint. Use the description of $\operatorname{ext}_A^h(M, L(j))$ via $\operatorname{grtop} P_h$ and the graded Nakayama lemma.

(4) Let $S = \bigoplus_{x=1}^n \Bbbk e_x$, let V be a finite-dimensional S-bimodule in degree 1, and let $A = T_S(V)$ be the tensor algebra. Prove that the multiplication map $V \otimes_S A \to A_+$ is an isomorphism of right A-modules, and deduce that S has the minimal graded projective resolution

$$0 \to (V \otimes_S A)(-1) \to A \to S \to 0,$$

so that A is Koszul.

Exercise 4.

(1) Let $P_\bullet \to M$ be a projective resolution with differentials d_h and $K_1 = \operatorname{Ker} d_0$. Prove that

$$\operatorname{Ext}^1_A(M,N) \cong \operatorname{coker}\big(\operatorname{Hom}_A(P_0,N) \to \operatorname{Hom}_A(K_1,N)\big),$$

where the map is restriction, and that $\operatorname{Ext}^h_A(M,N) \cong \operatorname{Ext}^{h-1}_A(K_1,N)$ for $h \geq 2$.

(2) Let $A = \Bbbk A_3/(e_{13})$ with $\deg e_{ij} = j - i$. Find the minimal graded projective resolution of L_1, and compute $\dim_\Bbbk \operatorname{ext}^h_A(L_1, L_y(j))$ for all h, y, j.

(3) Let A be as in Lecture 5. Prove that $\operatorname{ext}^1_A(S, S(j)) \cong \operatorname{Hom}_S\big((A_+/A_+^2)_{-j}, S\big)$ for all $j \in \mathbb{Z}$.

(4) Let A be as in (3). Prove that A is generated in degree 1 over S if and only if $\operatorname{ext}^1_A(S, S(j)) = 0$ for all $j \neq -1$.

Hint. You may use the isomorphism in (3).

(5) Let A be a graded algebra with $A_0 = \Bbbk$, locally finite-dimensional, and let M be a bounded below locally finite-dimensional graded right A-module whose minimal graded projective resolution $P_\bullet \to M$ has $P_h = 0$ for $h \gg 0$. Prove that

$$h_M(t) = h_A(t) \cdot \sum_{h \geq 0} (-1)^h \sum_{i \in \mathbb{Z}} \dim_\Bbbk \operatorname{ext}^h_A\big(M, \Bbbk(-i)\big)\, t^i.$$

Exercise 5.

(1) Let $A = \Bbbk\langle x, y\rangle/(yx,\ x^2 + y^2)$ with $\deg x = \deg y = 1$, a quadratic algebra. Show that $x^3 = 0$ and $y^3 = 0$, and deduce that $A_n = 0$ for $n \geq 3$ and $h_A(t) = 1 + 2t + 2t^2$.
Hint. $x^3 = x \cdot x^2 = -xy^2$ and $x^3 = x^2 \cdot x = -y^2x = -y(yx)$, and similarly for the other words of length 3.

(2) Let A be a graded algebra with $A_0 = \Bbbk$ and $h_A(t)$ a polynomial, and suppose that A is Koszul, with linear resolution $P_h = (T_h \otimes_\Bbbk A)(-h)$ of $\Bbbk$ and $b_h = \dim_\Bbbk T_h$. Prove that

$$\Big(\sum_{h\geq 0}(-1)^h b_h t^h\Big)\, h_A(t) = 1.$$

(3) Let A be as in (1). Using the identity of (2), prove that A is not Koszul, although it is quadratic.
Hint. Expand $1/h_A(t)$ to degree 4.

(4) Let $A = \Bbbk[x]/(x^3)$ with $\deg x = 1$. Find the minimal graded projective resolution of $\Bbbk$, and show that P_h is generated in degree $\lfloor 3h/2 \rfloor$ for all $h \geq 0$. Deduce that A is not Koszul. Is A quadratic?

(5) Let $S = \bigoplus_{y=1}^{n} \Bbbk e_y$, let V be a finite-dimensional bimodule in degree 1, and let $A = T_S(V)/(V \otimes_S V) = S \oplus V$. Prove that the minimal graded projective resolution of S has $P_h \cong (V^{\otimes_S h} \otimes_S A)(-h)$ for all $h \geq 0$, and deduce that A is Koszul.
Hint. Show that $\operatorname{Ker} d_0 = V$ and that $VA_+ = 0$, so each kernel is killed by A_+.